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Roll Number: 24102B0066

Branch: CMPN B

Cloud Computing Practical Examination

Total Marks: 40

Duration: 60 Minutes Instructions

- Perform all tasks on your own cloud virtual machine.
 - You may use AWS EC2, GCP Compute Engine VM, or Azure Virtual Machine.
 - Attach screenshots as evidence of completion.
 - Create a single PDF containing all screenshots and answers.
 - Mention question numbers clearly.
 - Ensure all screenshots clearly show successful execution of commands.
 - Look at the end of the pdf for submission instructions
-

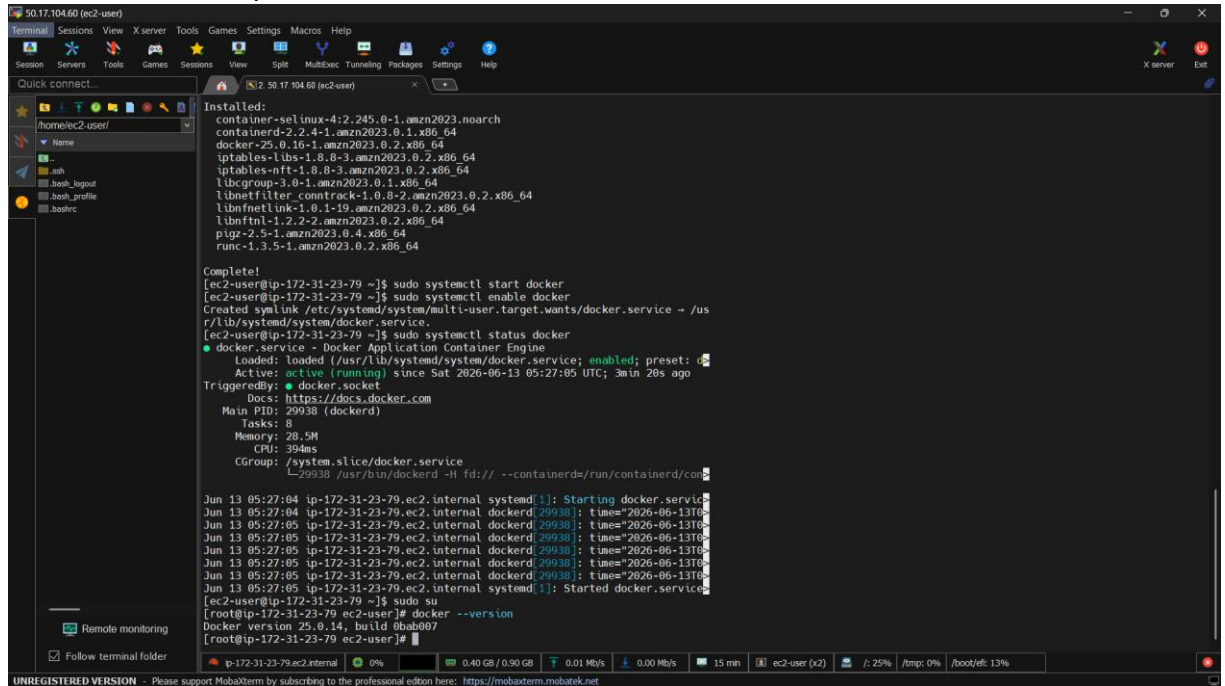
Question 1: Docker Installation & Nginx Deployment

Task

1. Install Docker on your cloud virtual machine.
2. Verify Docker installation.
3. Download the Nginx image.
4. Run the Nginx container.
5. Access the Nginx welcome page from your browser.

Required Screenshots

- Docker version output

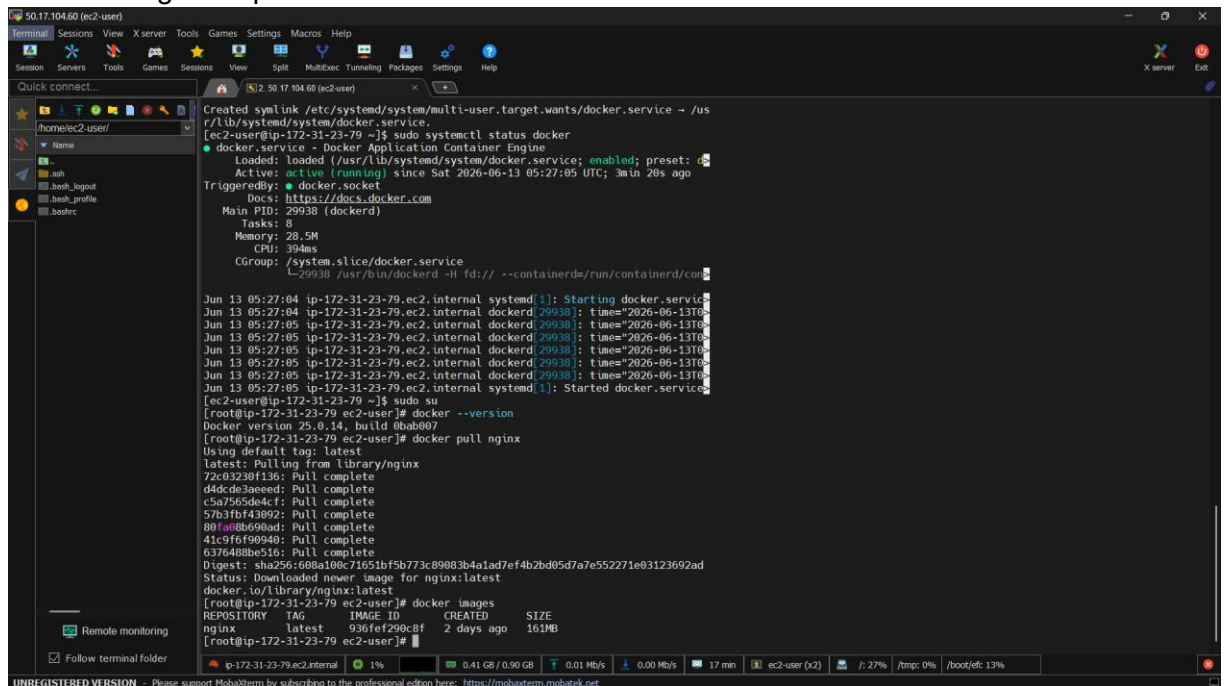


The screenshot shows a terminal window with the following content:

```
50.17.104.60 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split Multitex Tunneling Packages Settings Help
Quick connect...
home/ec2-user/
Name
ssh
ssh_logout
ssh_profile
bashrc
Installed:
container-engine-4:2.245.0-1.amzn2023.noarch
containerd-2.2.4-1.amzn2023.0.1.x86_64
docker-25.0.16-1.amzn2023.0.2.x86_64
iptables-libs-1.8.8-3.amzn2023.0.2.x86_64
iptables-nft-1.8.8-3.amzn2023.0.2.x86_64
libcgroup-3.0-1.amzn2023.0.1.x86_64
libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64
libnetfilter_link-1.0.1-19.amzn2023.0.2.x86_64
libnftnl-1.2.2-2.amzn2023.0.2.x86_64
pigz-2.5-1.amzn2023.0.4.x86_64
runc-1.3.5-1.amzn2023.0.2.x86_64
Complete!
[ec2-user@ip-172-31-23-79 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-23-79 ~]$ sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[ec2-user@ip-172-31-23-79 ~]$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset:
   Active: active (running) since Sat 2026-06-13 05:27:05 UTC; 3min 20s ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 29938 (dockerd)
      Tasks: 8
     Memory: 28.5M
        CPU: 394ms
     CGroup: /system.slice/docker.service
             └─29938 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/cont

Jun 13 05:27:04 ip-172-31-23-79.ec2.internal systemd[1]: Starting docker.servi
Jun 13 05:27:04 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T0
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T0
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T0
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Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T0
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T0
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal systemd[1]: Started docker.servic
[ec2-user@ip-172-31-23-79 ~]$ sudo su
[root@ip-172-31-23-79 ec2-user]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-172-31-23-79 ec2-user]#
```

- Docker images output



The screenshot shows a terminal window with the following content:

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50.17.104.60 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split Multitex Tunneling Packages Settings Help
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Name
ssh
ssh_logout
ssh_profile
bashrc
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[ec2-user@ip-172-31-23-79 ~]$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset:
   Active: active (running) since Sat 2026-06-13 05:27:05 UTC; 3min 20s ago
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     Docs: https://docs.docker.com
    Main PID: 29938 (dockerd)
      Tasks: 8
     Memory: 28.5M
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     CGroup: /system.slice/docker.service
             └─29938 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/cont

Jun 13 05:27:04 ip-172-31-23-79.ec2.internal systemd[1]: Starting docker.servi
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Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T0
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T0
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T0
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal systemd[1]: Started docker.servic
[ec2-user@ip-172-31-23-79 ec2-user]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-172-31-23-79 ec2-user]# docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
72c0320f136: Pull complete
d4dcde3aeeed: Pull complete
c5a7565de4cf: Pull complete
57b3bf430992: Pull complete
80fa086698ad: Pull complete
41c9f6f99940: Pull complete
6376488be516: Pull complete
Digest: sha256:688a108c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[ec2-user@ip-172-31-23-79 ec2-user]# docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx latest 936fef290c8f 2 days ago 161MB
[root@ip-172-31-23-79 ec2-user]#
```

- Docker ps output showing Nginx running

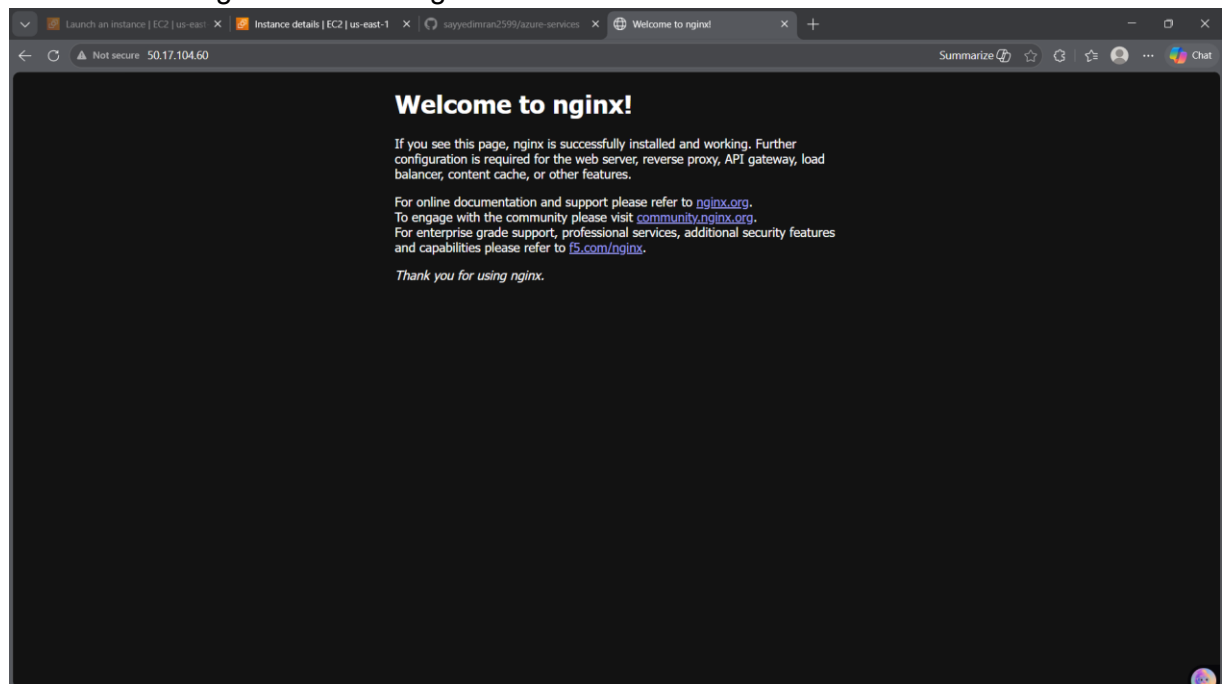
```

Jun 13 05:27:04 ip-172-31-23-79.ec2.internal systemd[1]: Starting docker.service
Jun 13 05:27:04 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T05:27:04.123456789Z" level=info msg="Starting dockerd"
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T05:27:05.123456789Z" level=info msg="Listening for connections"
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T05:27:05.123456789Z" level=info msg="API endpoint: http://localhost:2375"
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T05:27:05.123456789Z" level=info msg="Default storage driver: overlay2"
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T05:27:05.123456789Z" level=info msg="Default graph driver: overlay2"
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal systemd[1]: Started docker.service

[ec2-user@ip-172-31-23-79 ~]$ sudo su
[root@ip-172-31-23-79 ec2-user]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-172-31-23-79 ec2-user]# docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
72e0230f13b: Pull complete
d4d6e3a0e0d: Pull complete
c5a7565de4cf: Pull complete
57b3fbf43092: Pull complete
80fa08b690ad: Pull complete
41c9f6f90940: Pull complete
6376488be516: Pull complete
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd95d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[root@ip-172-31-23-79 ec2-user]# docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx latest 936fef290c8f 2 days ago 161MB
[root@ip-172-31-23-79 ec2-user]# docker run -d -p 80:80 --name mynginx nginx
61cd6eafc1d8e462690aaf5aa49a38d75a16f2a911b54e2a1d5ad87d853e252
[root@ip-172-31-23-79 ec2-user]# docker ps
CONTAINER ID IMAGE COMMAND NAMES CREATED STATUS
61cd6eafc1d8 nginx "/docker-entrypoint..." 8 seconds ago Up 7 seconds
0.0.0.0:80->80/tcp, :::80->80/tcp mynginx
[root@ip-172-31-23-79 ec2-user]#

```

- Browser showing "Welcome to nginx!"



Hint

Relevant Docker topics:

- Images
- Containers
- Port Mapping
- Container Lifecycle

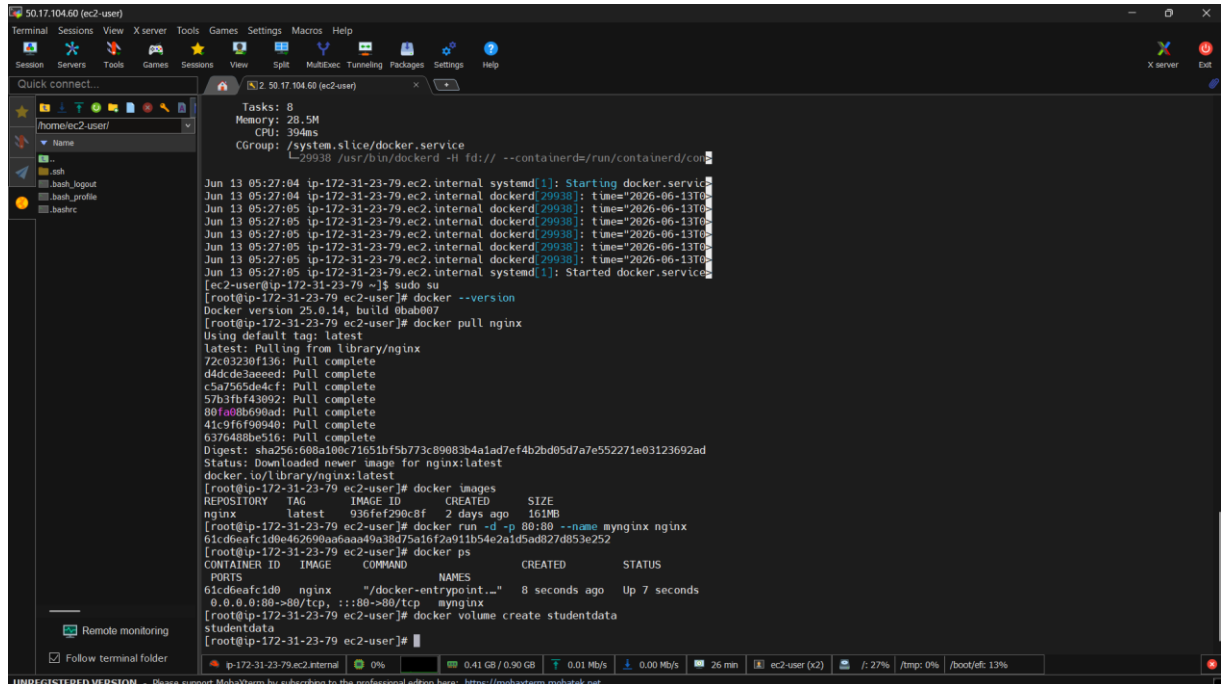
Question 2: Docker Volume Practical

Task

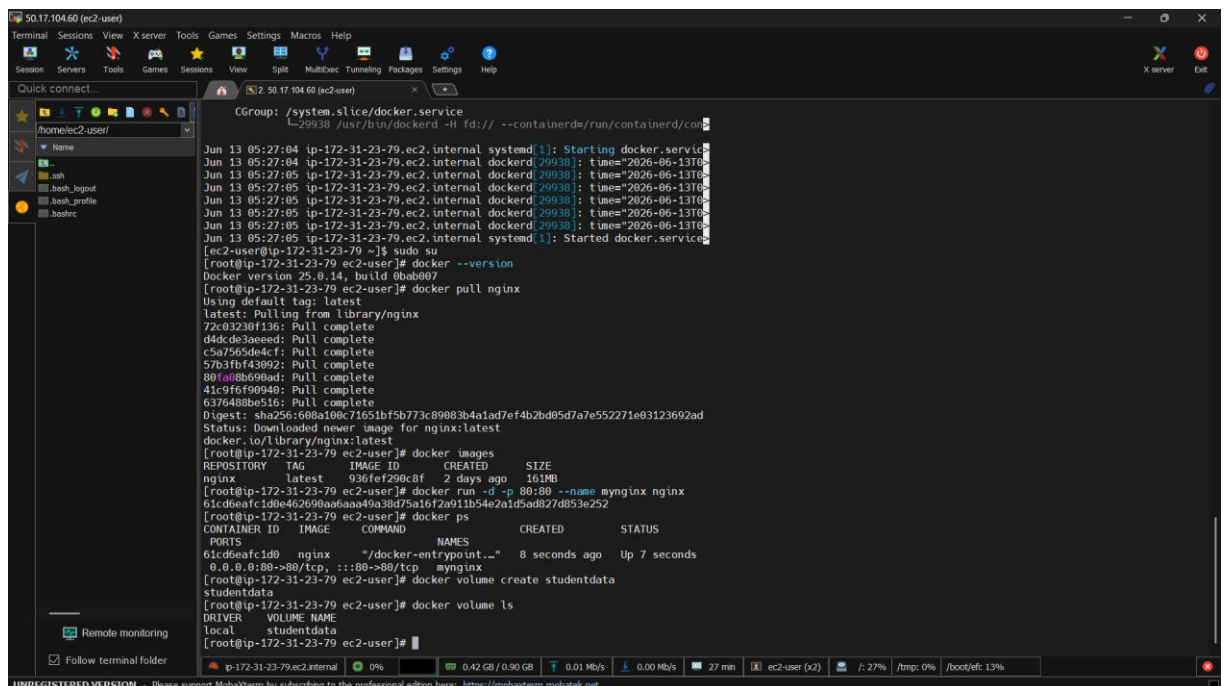
1. Create a Docker volume named: studentdata
2. Run an Ubuntu container using the volume.
3. Create a file: /data/info.txt
4. Write your name and roll number inside the file.
5. Display the contents of the file.

Required Screenshots

- Volume creation

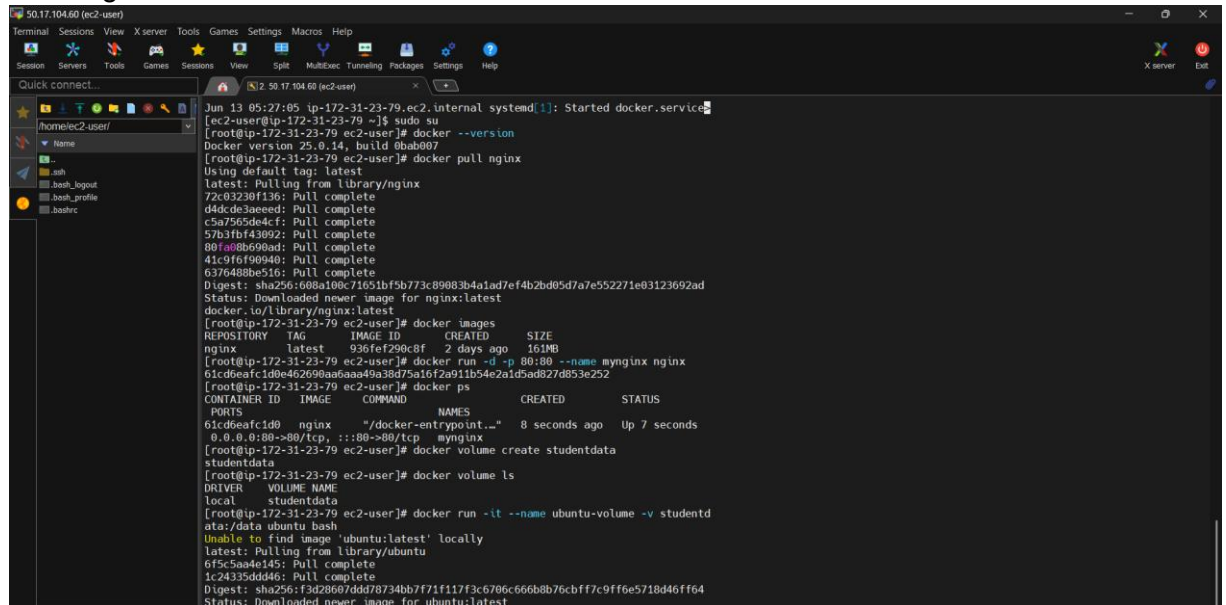


```
Jun 13 05:27:04 ip-172-31-23-79.ec2.internal systemd[1]: Starting docker.service
Jun 13 05:27:04 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T05:27:04.123456789Z" level=info msg="Starting Docker daemon"
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T05:27:05.123456789Z" level=info msg="Listening for connections"
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T05:27:05.123456789Z" level=info msg="API endpoint: http://0.0.0.0:2375"
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T05:27:05.123456789Z" level=info msg="Listening for connections"
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal systemd[1]: Started docker.service.
[ec2-user@ip-172-31-23-79 ~]$ sudo su
[root@ip-172-31-23-79 ec2-user]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-172-31-23-79 ec2-user]# docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
72c03230f136: Pull complete
d4dcde3aeeed: Pull complete
c5a7565de4cf: Pull complete
57b3fbf43092: Pull complete
80fa08b690ad: Pull complete
41c9f6f90940: Pull complete
6376488be516: Pull complete
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[root@ip-172-31-23-79 ec2-user]# docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx latest 936fef290c8f 2 days ago 161MB
[root@ip-172-31-23-79 ec2-user]# docker run -d -p 80:80 --name mynginx nginx
61cd6eafc1d0e462690aa6aaa49a38d75a16f2a911b54e2a1d5ad827d853e252
[root@ip-172-31-23-79 ec2-user]# docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS
61cd6eafc1d0 nginx "/docker-entrypoint..." 8 seconds ago Up 7 seconds
0.0.0.0:80->80/tcp, :::80->80/tcp mynginx
[root@ip-172-31-23-79 ec2-user]# docker volume create studentdata
studentdata
[root@ip-172-31-23-79 ec2-user]#
```



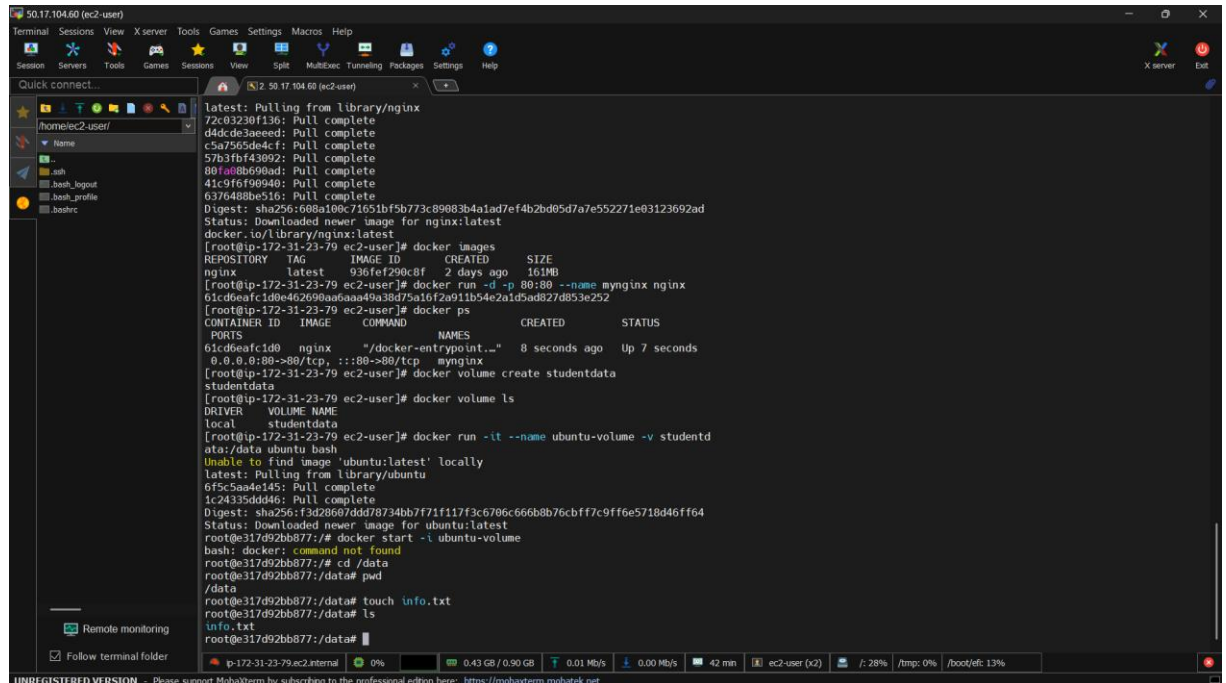
```
Jun 13 05:27:04 ip-172-31-23-79.ec2.internal systemd[1]: Starting docker.service
Jun 13 05:27:04 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T05:27:04.123456789Z" level=info msg="Starting Docker daemon"
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T05:27:05.123456789Z" level=info msg="Listening for connections"
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T05:27:05.123456789Z" level=info msg="API endpoint: http://0.0.0.0:2375"
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal dockerd[29938]: time="2026-06-13T05:27:05.123456789Z" level=info msg="Listening for connections"
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal systemd[1]: Started docker.service.
[ec2-user@ip-172-31-23-79 ~]$ sudo su
[root@ip-172-31-23-79 ec2-user]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-172-31-23-79 ec2-user]# docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
72c03230f136: Pull complete
d4dcde3aeeed: Pull complete
c5a7565de4cf: Pull complete
57b3fbf43092: Pull complete
80fa08b690ad: Pull complete
41c9f6f90940: Pull complete
6376488be516: Pull complete
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[root@ip-172-31-23-79 ec2-user]# docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx latest 936fef290c8f 2 days ago 161MB
[root@ip-172-31-23-79 ec2-user]# docker run -d -p 80:80 --name mynginx nginx
61cd6eafc1d0e462690aa6aaa49a38d75a16f2a911b54e2a1d5ad827d853e252
[root@ip-172-31-23-79 ec2-user]# docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS
61cd6eafc1d0 nginx "/docker-entrypoint..." 8 seconds ago Up 7 seconds
0.0.0.0:80->80/tcp, :::80->80/tcp mynginx
[root@ip-172-31-23-79 ec2-user]# docker volume create studentdata
studentdata
[root@ip-172-31-23-79 ec2-user]# docker volume ls
DRIVER VOLUME_NAME
local studentdata
[root@ip-172-31-23-79 ec2-user]#
```

- Running container



```
50.17.104.60 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Quick connect...
/home/ec2-user/
Name
.ssh
.ssh_logout
.ssh_profile
.sshrc
Jun 13 05:27:05 ip-172-31-23-79.ec2.internal systemd[1]: Started docker.service.
[ec2-user@ip-172-31-23-79 ~]$ sudo su
[root@ip-172-31-23-79 ec2-user]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-172-31-23-79 ec2-user]# docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
72c93230f136: Pull complete
04dcde3aeed: Pull complete
c5a7565de4cf: Pull complete
57b3fbf43092: Pull complete
80fa08b690ad: Pull complete
41c9f6f99948: Pull complete
6376d80be516: Pull complete
Digest: sha256:688a108c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e83123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[root@ip-172-31-23-79 ec2-user]# docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx latest 936fef290c8f 2 days ago 161MB
[root@ip-172-31-23-79 ec2-user]# docker run -d -p 80:80 --name mynginx nginx
61cd6eafc1d0e462690a6a9a38d75a16f2a911b54e2a1d5ad827d853e252
[root@ip-172-31-23-79 ec2-user]# docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS
PORTS
61cd6eafc1d0 nginx "/docker-entrypoint..." 8 seconds ago Up 7 seconds
0.0.0.0:80->80/tcp, :::80->80/tcp mynginx
[root@ip-172-31-23-79 ec2-user]# docker volume create studentdata
studentdata
[root@ip-172-31-23-79 ec2-user]# docker volume ls
DRIVER VOLUME NAME
local studentdata
[root@ip-172-31-23-79 ec2-user]# docker run -it --name ubuntu-volume -v studentd
ata:/data ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
6f5c5aa4e145: Pull complete
1c24335dd446: Pull complete
Digest: sha256:f3d28687d4d78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@e317d92bb877:/# docker start -i ubuntu-volume
bash: docker: command not found
root@e317d92bb877:/# cd /data
root@e317d92bb877:/data# pwd
/data
root@e317d92bb877:/data# touch info.txt
root@e317d92bb877:/data# ls
info.txt
root@e317d92bb877:/data#
```

- File creation



```
50.17.104.60 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Quick connect...
/home/ec2-user/
Name
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.ssh_logout
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04dcde3aeed: Pull complete
c5a7565de4cf: Pull complete
57b3fbf43092: Pull complete
80fa08b690ad: Pull complete
41c9f6f99948: Pull complete
6376d80be516: Pull complete
Digest: sha256:688a108c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e83123692ad
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61cd6eafc1d0e462690a6a9a38d75a16f2a911b54e2a1d5ad827d853e252
[root@ip-172-31-23-79 ec2-user]# docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS
PORTS
61cd6eafc1d0 nginx "/docker-entrypoint..." 8 seconds ago Up 7 seconds
0.0.0.0:80->80/tcp, :::80->80/tcp mynginx
[root@ip-172-31-23-79 ec2-user]# docker volume create studentdata
studentdata
[root@ip-172-31-23-79 ec2-user]# docker volume ls
DRIVER VOLUME NAME
local studentdata
[root@ip-172-31-23-79 ec2-user]# docker run -it --name ubuntu-volume -v studentd
ata:/data ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
6f5c5aa4e145: Pull complete
1c24335dd446: Pull complete
Digest: sha256:f3d28687d4d78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@e317d92bb877:/# docker start -i ubuntu-volume
bash: docker: command not found
root@e317d92bb877:/# cd /data
root@e317d92bb877:/data# pwd
/data
root@e317d92bb877:/data# touch info.txt
root@e317d92bb877:/data# ls
info.txt
root@e317d92bb877:/data#
```

- File contents displayed using cat

```

50.17.104.60 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split Multitex Tunneling Packages Settings Help
Quick connect...
/home/ec2-user/
Name
ssh
ssh_logout
ssh_profile
sshrc
88fa88b698ad: Pull complete
41c9f6f99940: Pull complete
6376488be516: Pull complete
Digest: sha256:688a108c71651bf5b773c89883b4a1ad7ef4b2bd05d7a7e552271e83123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[root@ip-172-31-23-79 ec2-user]# docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx latest 936fef298c8f 2 days ago 161MB
[root@ip-172-31-23-79 ec2-user]# docker run -d -p 80:80 --name mynginx nginx
61cd6eafc1d8e462699aa6aa49a38d75a16f2a911b54e2e1d5ad827d853e252
[root@ip-172-31-23-79 ec2-user]# docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS
61cd6eafc1d8 nginx "/docker-entrypoint..." 8 seconds ago Up 7 seconds
0.0.0.0:80->80/tcp, :::80->80/tcp mynginx
[root@ip-172-31-23-79 ec2-user]# docker volume create studentdata
studentdata
[root@ip-172-31-23-79 ec2-user]# docker volume ls
DRIVER VOLUME NAME
local studentdata
[root@ip-172-31-23-79 ec2-user]# docker run -it --name ubuntu-volume -v studentdata:/data ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
6f5c5aa4e145: Pull complete
1c24335dd446: Pull complete
Digest: sha256:f3d28687ddd8734bb7f71f117f3c6786c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@e317d92bb877:/# docker start -i ubuntu-volume
bash: docker: command not found
root@e317d92bb877:/# cd /data
root@e317d92bb877:/data# pwd
/data
root@e317d92bb877:/data# touch info.txt
root@e317d92bb877:/data# ls
info.txt
root@e317d92bb877:/data# echo "Name: Jai Vinayak Chavan" > info.txt
root@e317d92bb877:/data# echo "Roll No: 2410280066" >> info.txt
root@e317d92bb877:/data# cat info.txt
Name: Jai Vinayak Chavan
Roll No: 2410280066
root@e317d92bb877:/data#
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```

Question 3: Build Your First Docker Image

Task

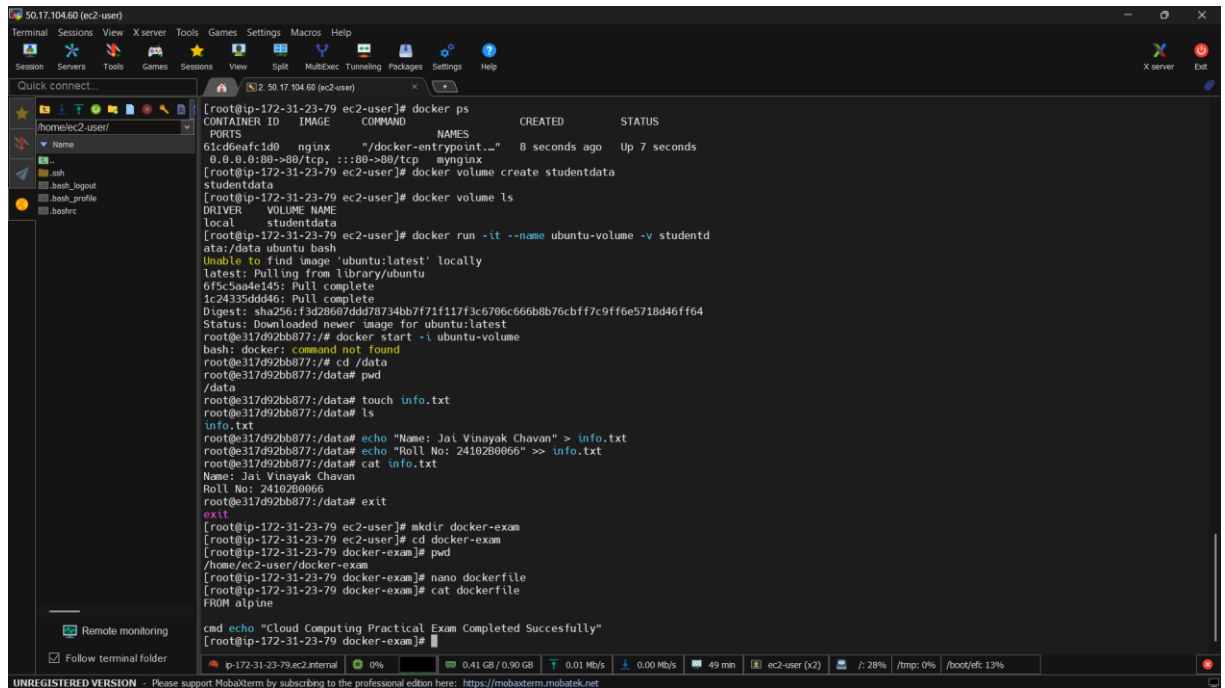
Create a custom Docker image that displays the following message when executed:

Cloud Computing Practical Exam Completed Successfully

Build the image and run it successfully.

Required Screenshots

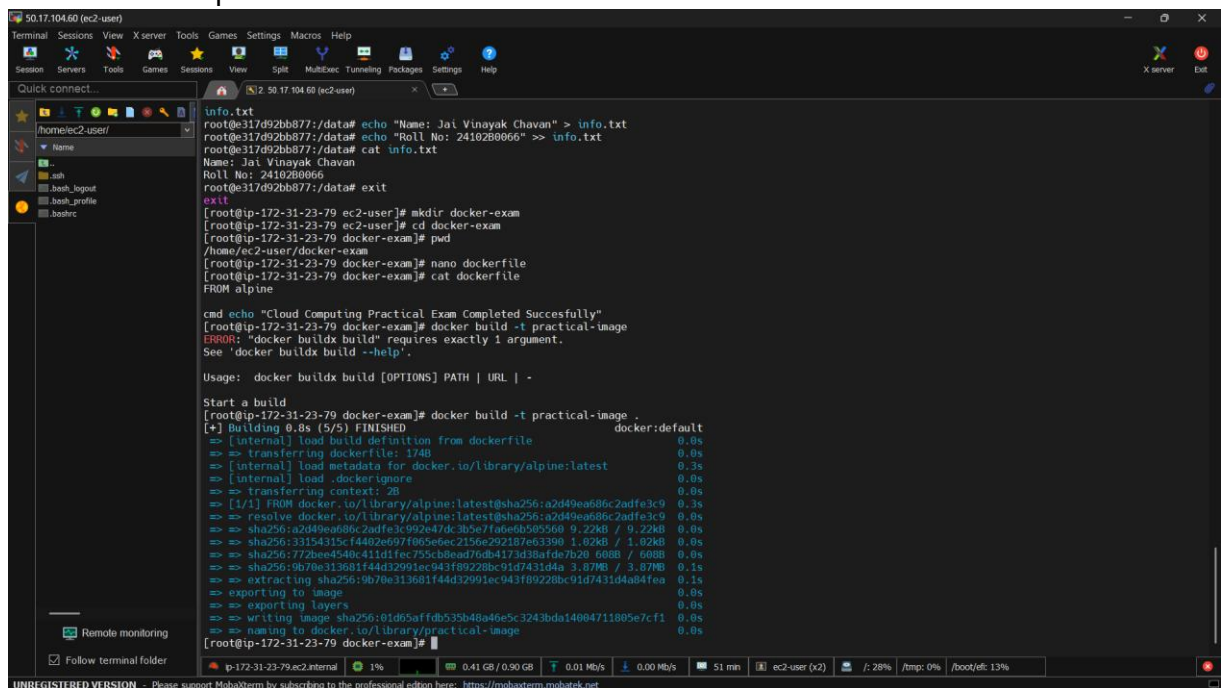
- Dockerfile



```
50.17.104.60 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split Multitex Tunneling Packages Settings Help
Quick connect...
home/ec2-user/
Name
ssh
bash_logout
bash_profile
bashrc
[root@ip-172-31-23-79 ec2-user]# docker ps
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS
61cd6eafcd9d        nginx              "/docker-entrypoint..." 8 seconds ago       Up 7 seconds        0.0.0.0:80->80/tcp, :::80->80/tcp
mynginx
[root@ip-172-31-23-79 ec2-user]# docker volume create studentdata
studentdata
[root@ip-172-31-23-79 ec2-user]# docker volume ls
DRIVER              VOLUME NAME
local               studentdata
[root@ip-172-31-23-79 ec2-user]# docker run -it --name ubuntu-volume -v studentdata:/data ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
6f5c5a4e145: Pull complete
1c2435dd4d46: Pull complete
Digest: sha256:f3d2b687dd5f8734bb7f71f11f73c6706c66db8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@e317d92bb877:/# docker start -i ubuntu-volume
bash: docker: command not found
root@e317d92bb877:/# cd /data
root@e317d92bb877:/data# pwd
/data
root@e317d92bb877:/data# touch info.txt
root@e317d92bb877:/data# ls
info.txt
root@e317d92bb877:/data# echo "Name: Jai Vinayak Chavan" > info.txt
root@e317d92bb877:/data# echo "Roll No: 2410280066" >> info.txt
root@e317d92bb877:/data# cat info.txt
Name: Jai Vinayak Chavan
Roll No: 2410280066
root@e317d92bb877:/data# exit
exit
[root@ip-172-31-23-79 ec2-user]# mkdir docker-exam
[root@ip-172-31-23-79 ec2-user]# cd docker-exam
[root@ip-172-31-23-79 docker-exam]# pwd
/home/ec2-user/docker-exam
[root@ip-172-31-23-79 docker-exam]# nano dockerfile
[root@ip-172-31-23-79 docker-exam]# cat dockerfile
FROM alpine

cmd echo "Cloud Computing Practical Exam Completed Successfully"
[root@ip-172-31-23-79 docker-exam]#
```

- docker build output



```
info.txt
root@e317d92bb877:/data# echo "Name: Jai Vinayak Chavan" > info.txt
root@e317d92bb877:/data# echo "Roll No: 2410280066" >> info.txt
root@e317d92bb877:/data# cat info.txt
Name: Jai Vinayak Chavan
Roll No: 2410280066
root@e317d92bb877:/data# exit
exit
[root@ip-172-31-23-79 ec2-user]# mkdir docker-exam
[root@ip-172-31-23-79 ec2-user]# cd docker-exam
[root@ip-172-31-23-79 docker-exam]# pwd
/home/ec2-user/docker-exam
[root@ip-172-31-23-79 docker-exam]# nano dockerfile
[root@ip-172-31-23-79 docker-exam]# cat dockerfile
FROM alpine

cmd echo "Cloud Computing Practical Exam Completed Successfully"
[root@ip-172-31-23-79 docker-exam]# docker build -t practical-image
ERROR: "docker buildx build" requires exactly 1 argument.
See 'docker buildx build --help'.

Usage: docker buildx build [OPTIONS] PATH | URL | -

Start a build
[root@ip-172-31-23-79 docker-exam]# docker build -t practical-image .
[*] Building 0.8s (5/5) FINISHED                                docker:default
=> [internal] load build definition from dockerfile               0.0s
=> => transferring dockerfile: 174B                               0.0s
=> [internal] load metadata for docker.io/library/alpine:latest  0.3s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [1/1] FROM docker.io/library/alpine:latest@sha256:a2d49ea686c2adfe3c9 0.3s
=> => resolve docker.io/library/alpine:latest@sha256:a2d49ea686c2adfe3c9 0.0s
=> => sha256:a2d49ea686c2adfe3c992e47dc3b5e7f1ae0b5095968 9.22kB / 9.22kB 0.0s
=> => sha256:33154315c14482e697f805e6ec2156e292187e63398 1.02kB / 1.02kB 0.0s
=> => sha256:772bee4540c411d1fec755cb8ead76db4173d38afde7b2b 608B / 608B 0.0s
=> => sha256:9b70e313681f44d32991ec943f89228bc91d7431d4a 3.87MB / 3.87MB 0.1s
=> => extracting sha256:9b70e313681f44d32991ec943f89228bc91d7431d4a84fea 0.1s
=> => exporting to image                                             0.0s
=> => exporting layers                                              0.0s
=> => writing image sha256:01d65affdb535b48a46e5c3243bdd14004711805e7cf1 0.0s
=> => naming to docker.io/library/practical-image                 0.0s
[root@ip-172-31-23-79 docker-exam]#
```

- docker images output

```
Roll No: 2410280066
root@ec317d92bb077:/data# exit
exit
[root@ip-172-31-23-79 ec2-user]# mkdir docker-exam
[root@ip-172-31-23-79 ec2-user]# cd docker-exam
[root@ip-172-31-23-79 docker-exam]# pwd
/home/ec2-user/docker-exam
[root@ip-172-31-23-79 docker-exam]# nano dockerfile
[root@ip-172-31-23-79 docker-exam]# cat dockerfile
FROM alpine

cmd echo "Cloud Computing Practical Exam Completed Successfully"
[root@ip-172-31-23-79 docker-exam]# docker build -t practical-image
ERROR: "docker buildx build" requires exactly 1 argument.
See "docker buildx build --help".

Usage:  docker buildx build [OPTIONS] PATH | URL | -

Start a build
[root@ip-172-31-23-79 docker-exam]# docker build -t practical-image .
[+] Building 0.8s (5/5) FINISHED          docker:default
=> [internal] load build definition from dockerfile          0.0s
=> => transferring dockerfile: 174B                          0.0s
=> [internal] load metadata for docker.io/library/alpine:latest 0.3s
=> [internal] load .dockerignore                             0.0s
=> => transferring context: 2B                                  0.0s
=> [1/1] FROM docker.io/library/alpine:latest@sha256:a2d49ea686c2adfe3c9 0.3s
=> => resolve docker.io/library/alpine:latest@sha256:a2d49ea686c2adfe3c9 0.0s
=> => sha256:a2d49ea686c2adfe3c92e47dc3b5e7f6e6b505560 9.22kB / 9.22kB 0.0s
=> => sha256:33154315c14482e697f065e6ec2156e292187e63390 1.02kB / 1.02kB 0.0s
=> => sha256:772bee4540c411d1fec755cb8ead76db4173d38afde7b20 608B / 608B 0.0s
=> => sha256:9b70e313681f44d32991ec943f89228bc91d7431d4a 3.87MB / 3.87MB 0.1s
=> => extracting sha256:9b70e313681f44d32991ec943f89228bc91d7431d4a84fea 0.1s
=> => exporting to image                                         0.0s
=> => exporting layers                                           0.0s
=> => writing image sha256:01d65affdb535b48a46e5c3243bda14004711805e7cf1 0.0s
=> => naming to docker.io/library/practical-image              0.0s
[root@ip-172-31-23-79 docker-exam]# docker images
REPOSITORY          TAG          IMAGE ID          CREATED          SIZE
nginx                latest      936fef298c8f      2 days ago      161MB
practical-image      latest      01d65affdb53      3 days ago      8.45MB
ubuntu               latest      30ba44506a6d      7 weeks ago     100MB
[root@ip-172-31-23-79 docker-exam]#
```

- docker run output

```
exit
[root@ip-172-31-23-79 ec2-user]# mkdir docker-exam
[root@ip-172-31-23-79 ec2-user]# cd docker-exam
[root@ip-172-31-23-79 docker-exam]# pwd
/home/ec2-user/docker-exam
[root@ip-172-31-23-79 docker-exam]# nano dockerfile
[root@ip-172-31-23-79 docker-exam]# cat dockerfile
FROM alpine

cmd echo "Cloud Computing Practical Exam Completed Successfully"
[root@ip-172-31-23-79 docker-exam]# docker build -t practical-image
ERROR: "docker buildx build" requires exactly 1 argument.
See "docker buildx build --help".

Usage:  docker buildx build [OPTIONS] PATH | URL | -

Start a build
[root@ip-172-31-23-79 docker-exam]# docker build -t practical-image .
[+] Building 0.8s (5/5) FINISHED          docker:default
=> [internal] load build definition from dockerfile          0.0s
=> => transferring dockerfile: 174B                          0.0s
=> [internal] load metadata for docker.io/library/alpine:latest 0.3s
=> [internal] load .dockerignore                             0.0s
=> => transferring context: 2B                                  0.0s
=> [1/1] FROM docker.io/library/alpine:latest@sha256:a2d49ea686c2adfe3c9 0.3s
=> => resolve docker.io/library/alpine:latest@sha256:a2d49ea686c2adfe3c9 0.0s
=> => sha256:a2d49ea686c2adfe3c92e47dc3b5e7f6e6b505560 9.22kB / 9.22kB 0.0s
=> => sha256:33154315c14482e697f065e6ec2156e292187e63390 1.02kB / 1.02kB 0.0s
=> => sha256:772bee4540c411d1fec755cb8ead76db4173d38afde7b20 608B / 608B 0.0s
=> => sha256:9b70e313681f44d32991ec943f89228bc91d7431d4a 3.87MB / 3.87MB 0.1s
=> => extracting sha256:9b70e313681f44d32991ec943f89228bc91d7431d4a84fea 0.1s
=> => exporting to image                                         0.0s
=> => exporting layers                                           0.0s
=> => writing image sha256:01d65affdb535b48a46e5c3243bda14004711805e7cf1 0.0s
=> => naming to docker.io/library/practical-image              0.0s
[root@ip-172-31-23-79 docker-exam]# docker images
REPOSITORY          TAG          IMAGE ID          CREATED          SIZE
nginx                latest      936fef298c8f      2 days ago      161MB
practical-image      latest      01d65affdb53      3 days ago      8.45MB
ubuntu               latest      30ba44506a6d      7 weeks ago     100MB
[root@ip-172-31-23-79 docker-exam]# docker run practical-image
Cloud Computing Practical Exam Completed Successfully
[root@ip-172-31-23-79 docker-exam]#
```

Hint

Relevant Dockerfile keywords:

- FROM
- CMD
- COPY

Think about:

- Which base image should be used?
- How does Docker know what command to execute when the container starts? •
- When would COPY be useful while building an image?

Bonus Challenge (Optional)

Explain in 2–3 lines each:

1. Difference between a Docker Image and a Docker Container.

-> **Docker Image:**

A Docker image is a read-only template that contains the application code, dependencies, libraries, and configuration required to run an application.

Docker Container:

A Docker container is a running instance of a Docker image that executes the application in an isolated environment.

2. Why are Docker Volumes used?

-> Docker volumes are used to store persistent data outside the container. The data remains available even if the container is stopped, deleted, or recreated.
